

1.1

Mějme komplexní vektorový
prostor V se skalárním
součinem $\langle \cdot \rangle$ a vektory
 $\vec{u}, \vec{v} \in V$

Počítáme:

$$\begin{aligned} & \|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = \\ &= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle \stackrel{\text{linearity}}{=} 2a \\ &= \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle \\ &\quad - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) = \end{aligned}$$

$$= 2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle \quad \text{symmetrie}$$

$$= 2\langle \vec{u}, \vec{v} \rangle + 2\overline{\langle \vec{u}, \vec{v} \rangle} = 4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

$$(\alpha + \bar{\alpha}) = 2\operatorname{Re}(\alpha) \quad \alpha \in \mathbb{C}$$

Üb 4:

$$\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

Ähnlich:

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 =$$

$$= \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle \quad \text{lin.}$$

$$= \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle -$$

$$- (\langle \vec{u}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle - i\langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) =$$

$$= -2i \langle \vec{u}, \vec{v} \rangle + 2i \langle \vec{v}, \vec{u} \rangle =$$

$$= -2i \left(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} \right) =$$

$$u \in \mathbb{C}$$

$$u - \bar{u} = 2\operatorname{Im}(u)$$

$$= -2i \cdot 2i \cdot \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) =$$

$$= 4 \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

Yedy máme:

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} \|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2$$
